General Mathematics — Exercise Sheet 13

Oriented geometry and work

Variant: Statements

Exercise 1 (An oriented triangle and its plane). The points $A = (1, -1, 2)$, $B = (3, 0, 1)$ and $C = (0, 2, 4)$ are given. Find the triangle's area and the unit normal selected by the ordered edges $\overrightarrow{AB}$ and $\overrightarrow{AC}$. Write a normal equation for its plane and explain what reverses when B and C are interchanged. For the separate plane $2x - y + 2z = 4$, find one point and two independent directions giving its parametric equations. Verify that the parameters describe every point of that plane.

Exercise 2 (Changing the reference point for torque). Use right-handed Cartesian axes. A force $\boldsymbol{F} = (1, 3, -2)$ newtons acts at position $\boldsymbol{r} = (2, -1, 1)$ metres relative to an origin O. Find its torque about O and about a new reference point O' whose position relative to O is $\boldsymbol{q} = (1, 1, 0)$ metres. Find the signed torque component along the positive third direction in each case. If the same constant force acts over displacement $\boldsymbol{d} = (-1, 2, 2)$ metres, compute its work and explain why shifting the reference point does not change that displacement or work.

Exercise 3 (Coplanarity and a volume constraint). Let $\boldsymbol{a} = (1, 2, 0)$, $\boldsymbol{b} = (0, 1, 1)$ and $\boldsymbol{c}(t) = (t, 1, 2)$ for $t \in \mathbb{R}$. Find the signed parallelepiped volume as a function of t. Find all parameters for which the vectors are coplanar and express $\boldsymbol{c}(t)$ as a combination of $\boldsymbol{a}$ and $\boldsymbol{b}$ at those parameters. Find all t for which the ordinary volume is at most 3. Finally, replace $\boldsymbol{b}$ by $(2, 4, 0)$ and state the signed volume for every t.

Exercise 4 (The vector triple-product identity). For arbitrary $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{R}^3$, prove by components

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b}).$$

Deduce the corresponding formula for $(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{c}$. No vector is assumed nonzero or independent.